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Introduction

- Sense of agency (SoA) refers to the feeling of being in control of one's own actions and their effects on external world.
- Gaze cueing (GC) refers to the human tendency to shift attention in the direction of another person's gaze; it is a fundamental social cue, allowing individuals to infer the attentional focus of others and engage in joint attention, a core component of human social interaction (Tomasello et al., 2005).
- While prior research has shown that gaze behavior can influence SoA (e.g., mutual gaze enhancing agency; Ulloa et al., 2019;), the reverse relationship has not been directly tested. Our study addresses this gap by investigating whether the internal state of SoA shapes the strength of GCE. Importantly, we keep an optimal time window of the internal state of SoA (e.g., Farrer et al, 2013) to assure the occurrence of the action-effect integration process.

Method

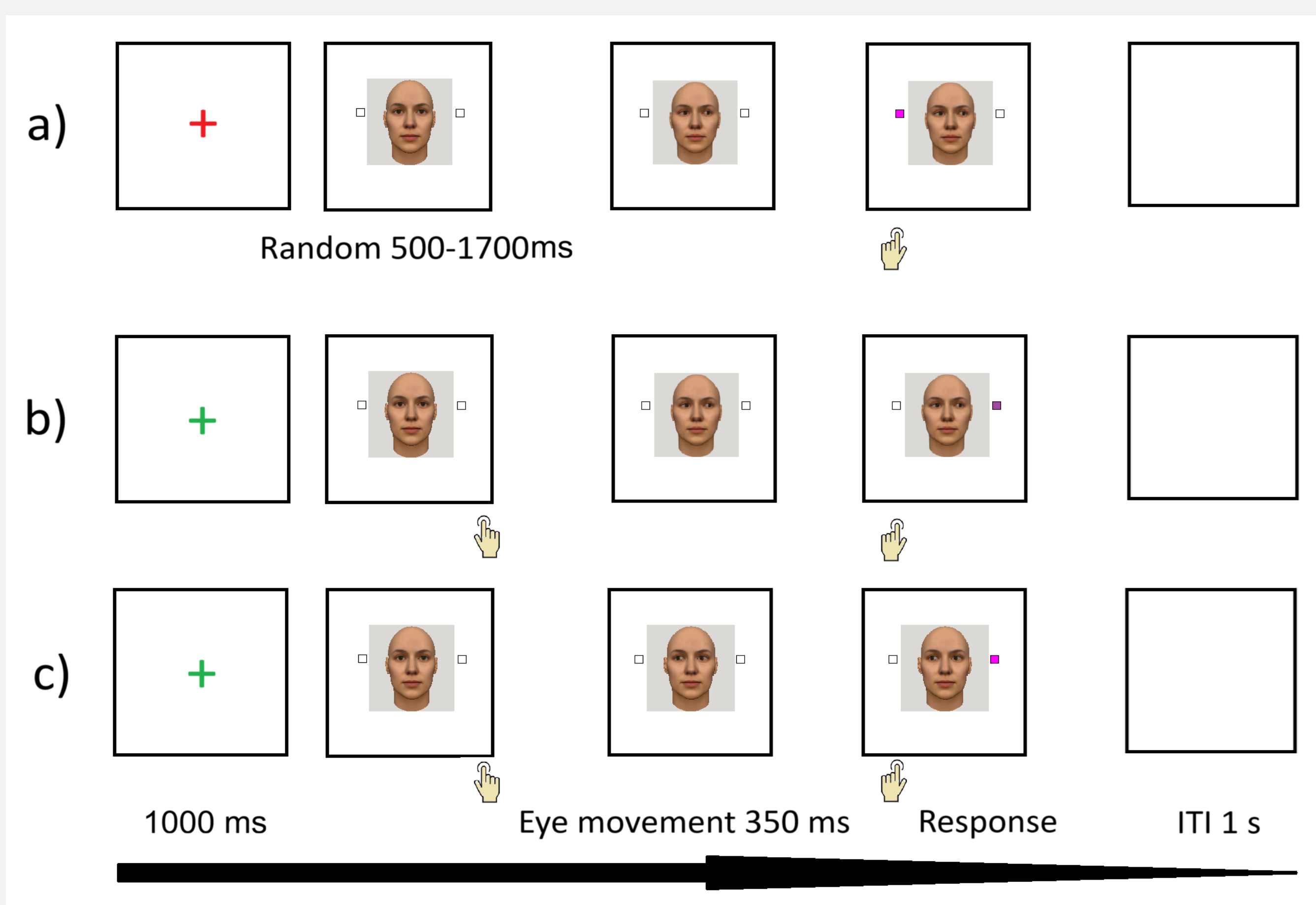
Participants 43 students participated in the experiment (7 males, age M(SD)=24.7(8.4) years old). All participants had normal or corrected to normal vision.

Design: 2 (Gaze Cueing: congruent vs. incongruent) x 3 (Action-Outcome Consistency: Consistent vs. Inconsistent vs. Control).

Procedure: 16 practice trials; 28 trials per condition (168 overall); Response buttons were counterbalanced across participants; E-prime controlled stimuli presentation and responses.

Task 1: Press Left/Right button ("4" vs "6", extended keyboard) of your choice if you see + (Right hand), don't press a button if you see +.

Task2: Color discrimination (Left hand, "a" vs "d"); violet (BlueViolet, RGB: 138, 43, 226) and magenta (Magenta, RGB: 255, 0, 255).

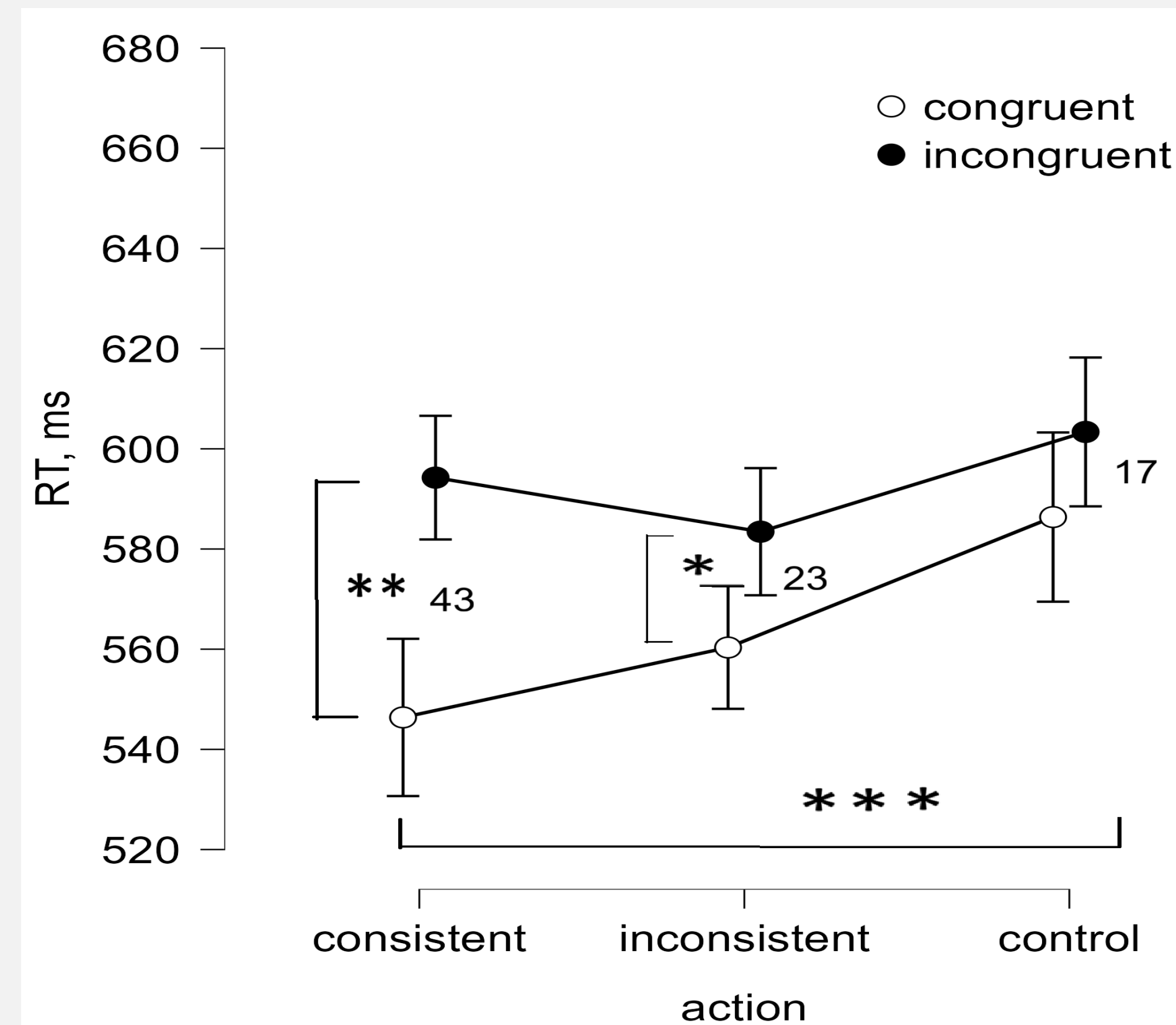


Trial timeline examples: a) Control condition (+ /NoGo), incongruent gaze cuing (gaze is away from the target/colored rectangle); b) congruent gaze cuing, c) incongruent gaze cuing. +/Go; ITI – intertrial interval.

References

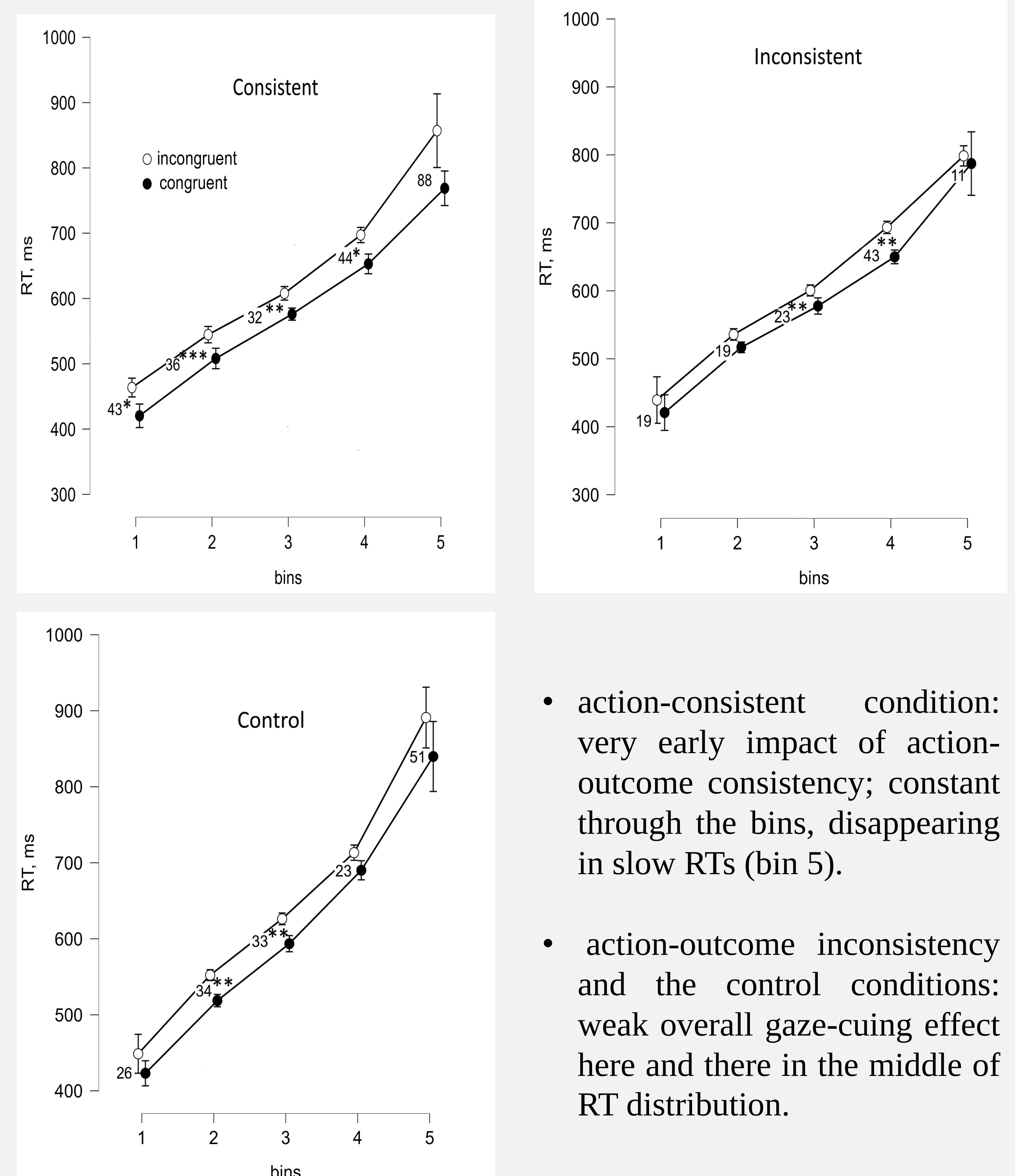
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Results



RT of correct responses was trimmed M±2SD per condition per participant (data of 24 were accepted). Data were averaged by subject, rANOVA was run. Adjusted Bonferroni post-hoc test applied. Interaction ($F(2,46)=3.82$, $p=0.029$, $\eta_p^2=0.14$).

Bin Analysis



- action-consistent condition: very early impact of action-outcome consistency; constant through the bins, disappearing in slow RTs (bin 5).
- action-outcome inconsistency and the control conditions: weak overall gaze-cuing effect here and there in the middle of RT distribution.

Conclusion

- The results unambiguously suggest that our participants did experience an internal state of agency which influenced the GCE.
- Note that an outcome of the action in our case is a 'doubled' one: immediately after a button press participants observe gaze movement and then a target appearance which was still within the action-effect time window (e.g., Volosin et al., 2025).
- We suppose that either the target appearance played an action-effect reinforcement role or SoA has a 'spillover' effect on processes if they start within the action-effect window (color discrimination in our case). We argue that SoA has a 'spillover' effect on processes if they start within the action-effect time window.
- Overall, the results suggest a mediating role of Sense of Agency in Gaze cuing effect which broadly connects consequences of our intentions and actions with a fundamental social cue.